Generative AI

(INF 426)

Assignment 2 – Report

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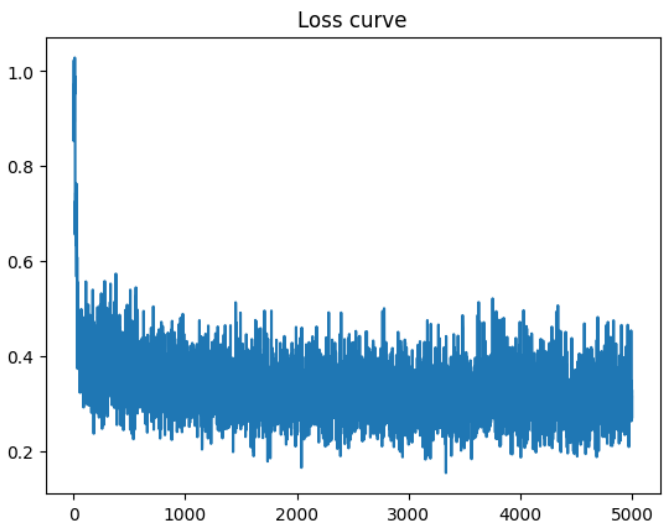
30/06/2025

Screenshots for Task 1: DDPM pipeline with Swiss-Roll

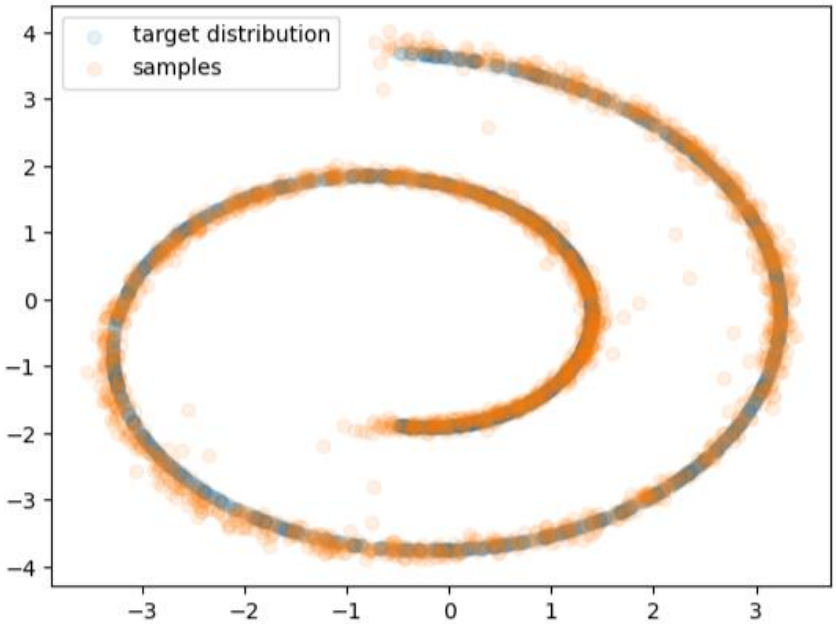
1. Training loss curve



2. Chamfer Distance reported after executing the Jupyter Notebook

```
DDPM Chamfer Distance: 19.1713
```

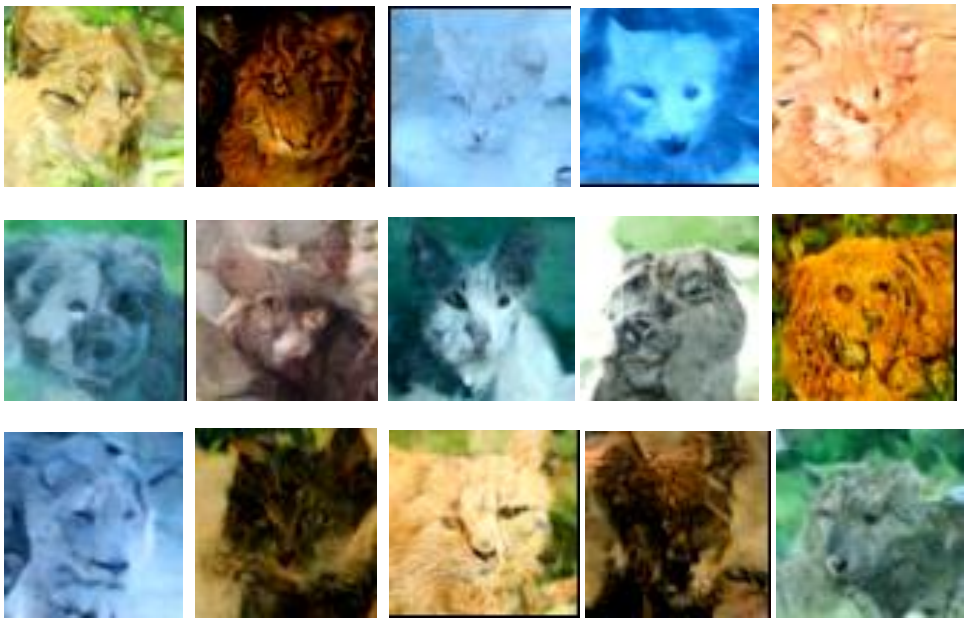
3. Visualization of the sampled particles



Task 2: Image Diffusion

Note: The number of training steps was reduced to 10.000, and the training took approximately 26 hours to complete. Additionally, the training was performed on a CPU due to the unavailability of a GPU.

Samples



FID Score

```
  warnings.warn(
100%|
100%|
FID: 65.6091222580734
```

Note: The afhq_inception_v3.ckpt file located in image_diffusion_todo/fid was removed to reduce the zip file size to under 10 MB for uploading to Moodle.